

ANCOVA: Analysis of Covariance

Hotheifa Al-3Assi

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Education is a vital part of our lives, helping us to understand everything and improve in all aspects. Therefore, developing the educational process and its methods is crucial.

This study aims to compare three teaching methods—traditional(Group1), online(Group2), and blended(Group3)—to determine if the teaching method significantly impacts students' final exam scores. Forty-Five students were randomly assigned to one of the three teaching methods. All students took a standardized final exam at the end of the course. Results from pre-tests, administered before the course began, were available and will be considered in the analysis.

Are there differences between teaching methods when controlling the pre-test?

We want to know if there is a difference between the three teaching methods?

Did the students' pre-test scores affect their final exam scores?

Did the teaching methods and pre-test results influence the final exam?

The Data:

First six Rows:

Table 1: The Data

Id	Group	Pre_test	Post_test
1	grp1	14.1	14.1
2	grp1	14.5	14.3
3	grp1	15.7	14.9
4	grp1	16.0	15.3
5	grp1	16.5	15.7
6	grp1	16.9	16.2

last six rows:

Table 2: The Data

Id	Group	Pre_test	Post_test
39	grp3	17.5	14.4
40	grp3	17.6	13.8
41	grp3	17.8	14.3
42	grp3	17.9	13.8
43	grp3	18.4	15.4
44	grp3	18.5	15.1
45	grp3	19.0	15.5

Dimensions of the Data set:

Table 3: Dimensions of the Dataset

Dimension	Value
Number of Rows	45
Number of Columns	3

Table 3 shows that the sample size is 45 and Variables we interest is 3: Group, Pre-test and Post-test.

Variable type :

Table 4: Structure of the Dataset

Variable	Type	Example
group	character	grp1,grp1,grp1
pretest	numeric	14.1,14.5,15.7
posttest	numeric	14.1,14.3,14.9

Table 4 shows that the variables pre-test,post-test is numerical and The variable Group is Categorical.

Descriptives Statistics For Numirical Variables :

- The Numerical Descriptive Statistics:

Table 5: Descriptives Statistics for Numirical Variable

	mean	sd	median	min	max	se
pretest	16.92	1.49	17.0	13.7	19.8	0.22
posttest	15.24	2.02	15.3	11.0	19.4	0.30

Table 5 shows that the mean of pre-test is 16.92 and the Standard deviation is 1.49 and the mean of post-test is 15.24 and Standard deviation 2.02.

- The Graphical Descriptive Statistics:

Figure 1: Boxplot of Numirical Variables

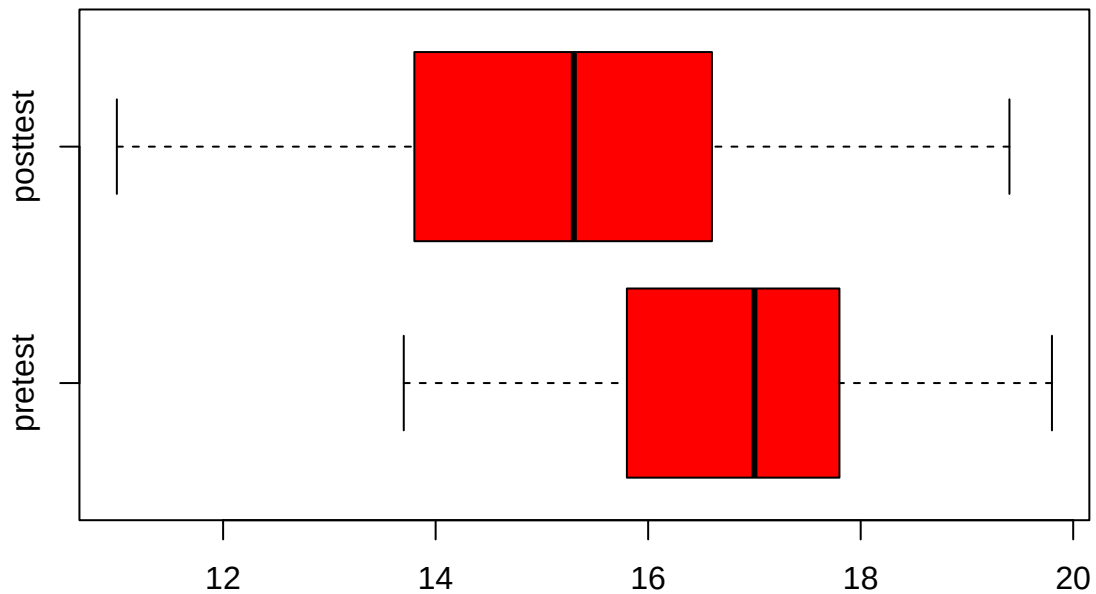


Figure 1 shows that the mean of Pre-Test is higher than the Mean of the Post-Test ,while the spread of Post-test is higher than the spread of Pre-Test.

Categorical Variables :

Traditional Method (Group1), Online Method(Group2), and Blended Method(Group3) All the Groups have the same size $n_i = 15$, $i = 1, 2, 3$.

Figure 2: Boxplot for Pre-Test with Groups

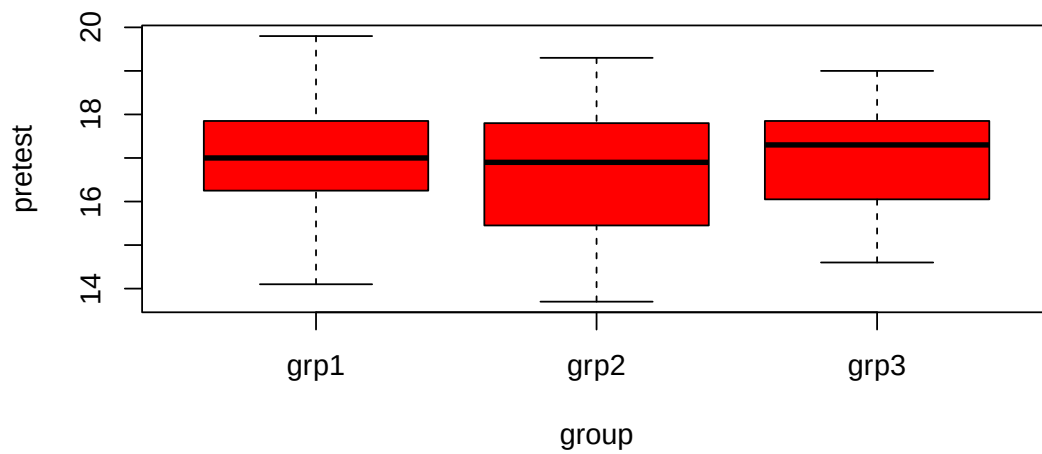


Figure 2 shows that the mean for each group in the pre-test is almost equal, while there is a small difference in spread.

Figure 3 : Boxplot for Post-Test with Groups

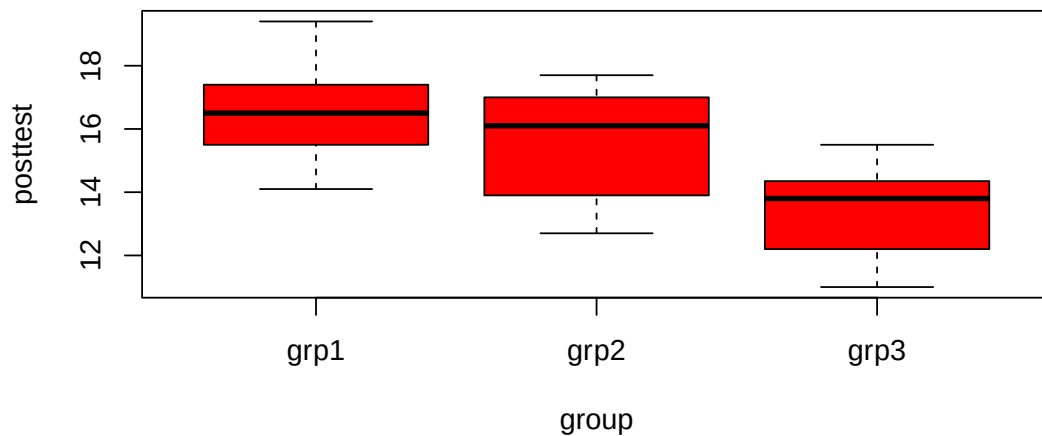


Figure 3 shows that the mean for each group in the pre-test is difference , while there is a small difference in spread.

Inference Statistics:

For inferential statistics, we used regression analysis where the response variable was the Post-test which is numerical and the predictors were the pre-test which is numerical and the Groups which is Categorical. The results given in Table 6.

Table 6: Regression Analysis

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-0.9409	0.7257	-1.2964	0.2021
pretest	1.0278	0.0420	24.4606	0.0000
group1grp2	-0.6411	0.1514	-4.2354	0.0001
group1grp3	-2.9846	0.1503	-19.8617	0.0000

The results Show that :

$\hat{B}_1 = 1.0278$, When the pre-test increases by one Grade, the post-test average increases by 1.0278 With other factors remaining constant.

$\hat{B}_2 = -0.6411$, When moving from the traditional teaching method to online teaching methods will lower the average of post-test by -0.6411, With other factors remaining constant.

$\hat{B}_3 = -2.9846$, When moving from the traditional teaching method to blended teaching methods will lower the average of post-test by -2.9846, With other factors remaining constant.

The Hypotheses:

Adjusted Post-Test:

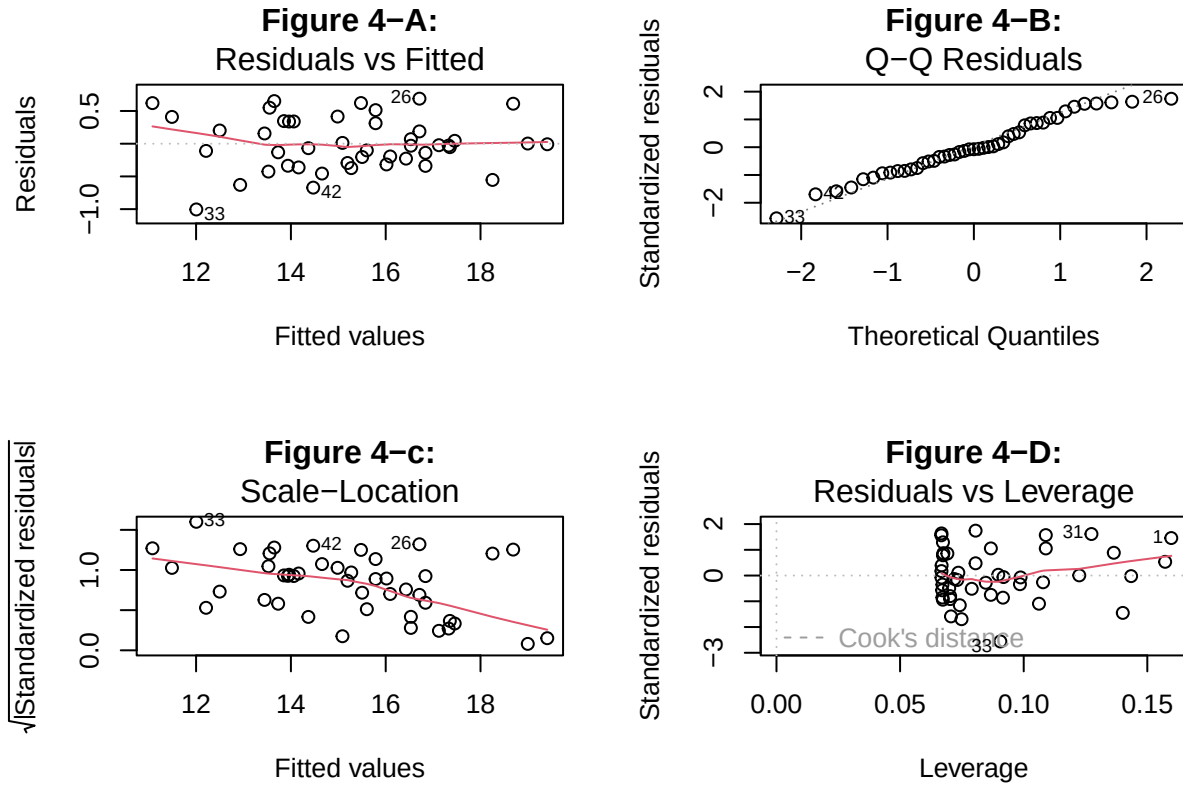
H_0 : There are no statistically significant differences between the adjusted post-test means of the three teaching methods (after adjusting for pre-test) between the groups.

H_1 : There is at least one statistically significant difference between the adjusted post-test means between the groups.

p -value for $\hat{\beta}_2 = 0.0001 < \alpha = 0.05$, p -value for $\hat{\beta}_3 = 0.0000 < \alpha = 0.05$, we reject H_0 at $\alpha = 0.05$, There is statistically significant difference between the adjusted post-test means between the groups.

check the assumption :

We will now verify the assumptions of the linear regression model shown above.
The results are shown in the figure:



- Figure 4-A: Residuals vs Fitted plot : The points appear scattered around zero; there is no specific pattern, and linearity is achieved.
- Figure 4-B: Q-Q Residuals plot : The points nearly lie on a straight line; the data follow a normal distribution.
- Figure 4-C: Scale-Location : The points appear scattered around zero; there is no specific pattern, Constant variance is achieved.
- Figure 4-D: Residuals vs Leverage : The diagnostic plot of residuals versus leverage indicated no highly influential observations, as no data points exceeded Cook's Distance thresholds.

Difference between the three teaching methods:

we used analysis of variance to find the difference between the three teaching methods, where the response is post-Test which is numerical and Groups which is categorical. The results given in Table 7.

Table 7: ANOVA Table

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
group1	2	72.13378	36.06689	13.99629	2e-05
Residuals	42	108.22933	2.57689	NA	NA

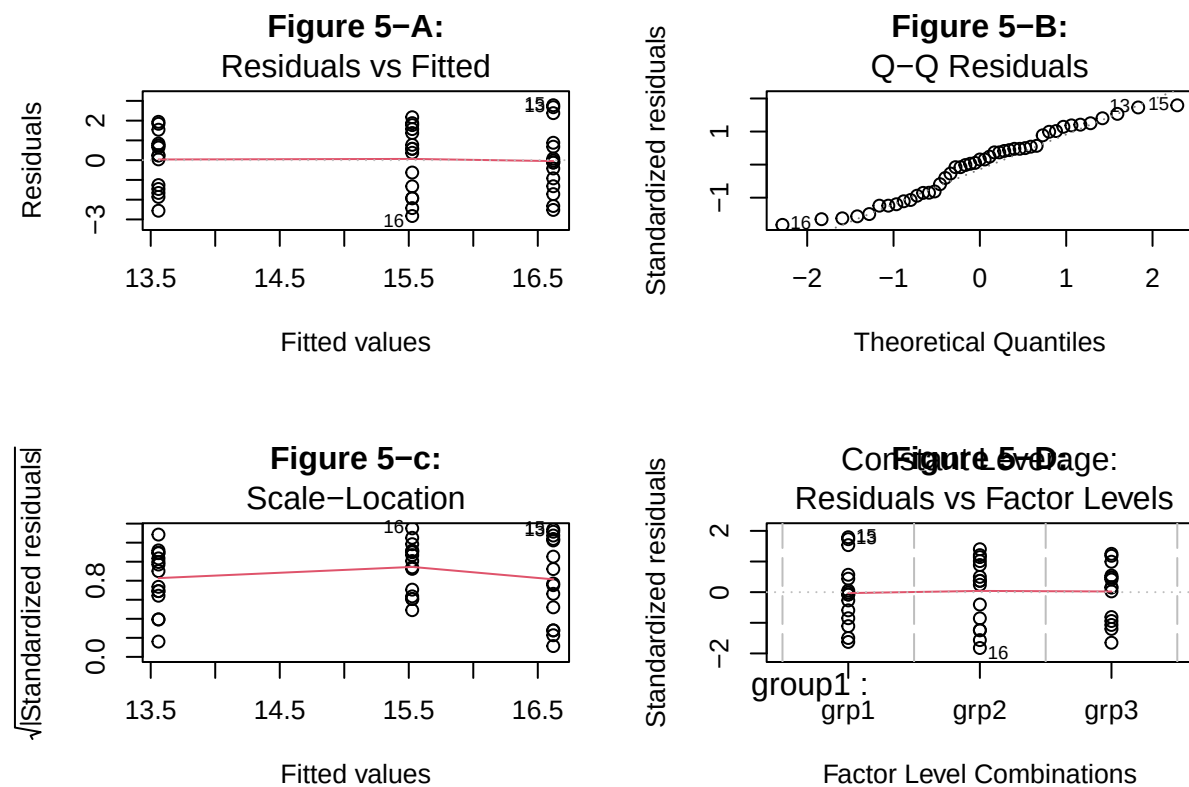
The Hypotheses:

H_0 : There is no statistically significant differences between means of the three teaching methods.

H_a : There is statistically significant differences between means of the three teaching methods.

$p\text{-value} = 0.00002 < \alpha = 0.05$, we reject H_0 at $\alpha = 0.05$, There is statistically significant differences between means of the three teaching methods.

check the assumption :



- Figure 5-A: Residuals vs Fitted plot : The points appear scattered around zero; there is no specific pattern.
- Figure 5-B: Q-Q Residuals plot : The points nearly lie on a straight line; the data follow a normal distribution.
- Figure 5-C: Scale-Location : The points appear scattered around zero; there is no specific pattern, Constant variance is achieved.
- Figure 5-D: Residuals vs Leverage : The diagnostic plot of residuals versus leverage indicated no highly influential observations, as no data points exceeded Cook's Distance thresholds.

Pairwise Comparisons:

Since we rejected the null hypothesis, this means that there are differences between the means. The comparison method was used Bonferroni. Table 8 shows the results:

Table 8: Pairwise Comparisons (Bonferroni Adjustment)

	grp1	grp2
grp2	0.2074	NA
grp3	0.0000	0.0051

The Hypotheses:

- Traditional Method (Group1) and Online Method(Group2).

$$H_0 : \mu_{Traditional} = \mu_{Online}.$$

$$H_1 : \mu_{Traditional} \neq \mu_{Online}.$$

$p - value = 0.2074 < \alpha = 0.05$, we do not reject H_0 at $\alpha = 0.05$, there is no difference between Traditional and Online Methods.

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- Traditional Method (Group1) and Blended Method(Group3).

$$H_0 : \mu_{Traditional} = \mu_{Blended}.$$

$$H_1 : \mu_{Traditional} \neq \mu_{Blended}.$$

$p - value = 0.0000 < \alpha = 0.05$, we reject H_0 at $\alpha = 0.05$, there is difference between Traditional and Blended Methods.

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- Online Method (Group2) and Blended Method(Group3).

$$H_0 : \mu_{Online} = \mu_{Blended}.$$

$$H_1 : \mu_{Online} \neq \mu_{Blended}.$$

$p - value = 0.0051 < \alpha = 0.05$, we reject H_0 at $\alpha = 0.05$, we reject H_0 at $\alpha = 0.05$, there is difference between Online and Blended Methods.

Pre-Test on Post-Test:

To Study the relationship between Pre-Test and Post-Test we used regression analysis where the response variable was the Post-test which is numerical and the predictors were the pre-test which is numerical. The results given in Table 9.

Table 9: Regression Analysis

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-1.8388	2.3589	-0.7795	0.4399
pretest	1.0094	0.1389	7.2658	0.0000

The results Show that :

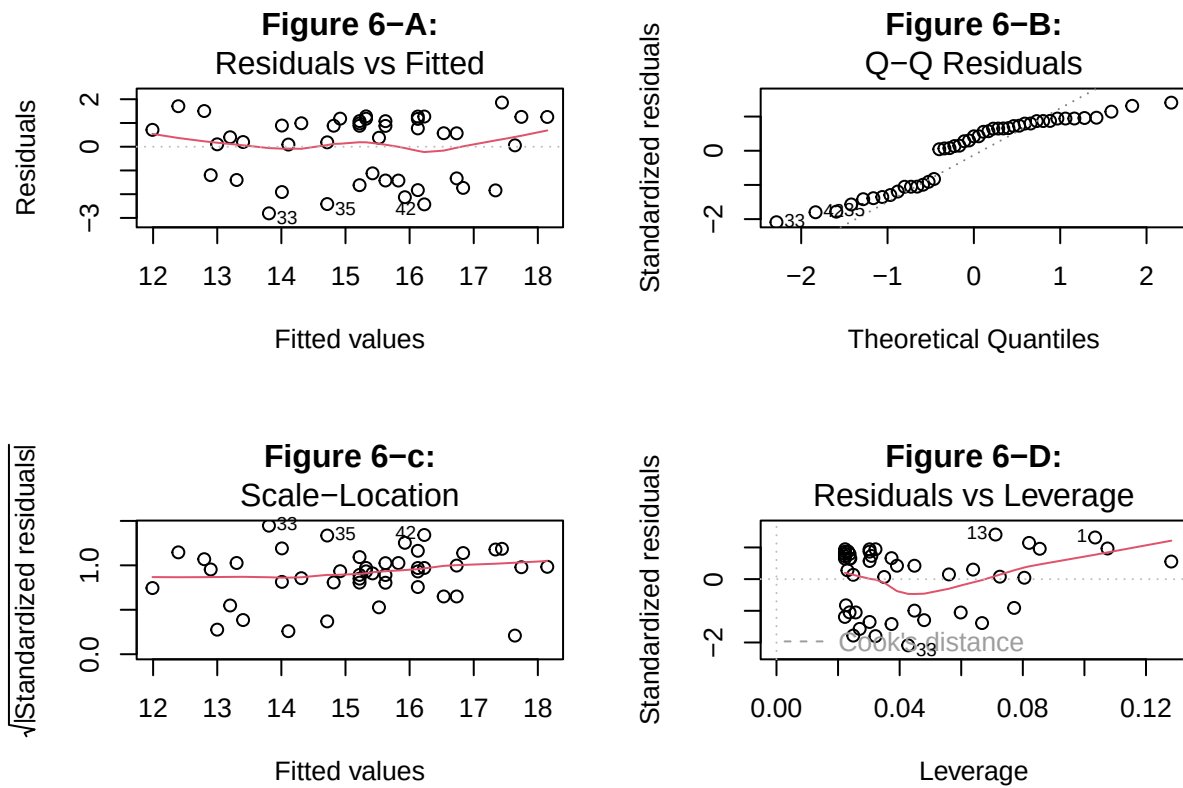
$\hat{B}_1 = 1.0094$, When the pre-test increases by one Grade, the post-test average increases by 1.0278.

The Hypotheses:

H_0 ::There is no statistically significant effect of pre-test on post-test. H_1 :There is statistically significant effect of pre-test on post-test.

p -value for $\hat{\beta}_2 = 0.0000 < \alpha = 0.05$,we reject H_0 at $\alpha = 0.05$,there is statistically significant effect of pre-test on post-test.

check the assumption :



- Figure 6-A: Residuals vs Fitted plot : The points appear scattered around zero; there is no specific pattern, and linearity is achieved.
- Figure 6-B: Q-Q Residuals plot : The points nearly lie on a straight line; the data follow a normal distribution.
- Figure 6-C: Scale-Location : The points appear scattered around zero; there is no specific pattern, Constant variance is achieved.
- Figure 4-D: Residuals vs Leverage : The diagnostic plot of residuals versus leverage indicated no highly influential observations, as no data points exceeded Cook's Distance thresholds.

Interaction:

For Interaction we used regression analysis where the response variable was the Post-test which is numerical and the predictors were the pre-test which is numerical and the Groups which is Categorical. The results given in Table 10.

Table 10: Regression Analysis With Interaction

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-0.6860	1.1840	-0.5794	0.5657
pretest	1.0128	0.0690	14.6783	0.0000
group1grp2	-1.3770	1.6857	-0.8169	0.4190
group1grp3	-2.9349	1.8730	-1.5670	0.1252
pretest:group1grp2	0.0438	0.0996	0.4400	0.6623
pretest:group1grp3	-0.0030	0.1095	-0.0273	0.9784

The results Show that :

$\hat{B}_1 = 1.0128$, When the pre-test increases by one unit, the post-test average increases by 1.0128 With other factors remaining constant.

$\hat{B}_2 = -1.3770$, when Using online teaching methods will decreases the average of post-test by -1.3770 With other factors remaining constant.

$\hat{B}_3 = -2.9349$, when Using blended teaching methods will decreases the average of post-test by -2.9349 With other factors remaining constant.

$\hat{B}_4 = 0.0438$, when the interaction between the pre-test and the group2 increases by one unit the Post-test average increase by 0.0438 With other factors remaining constant.

$\hat{B}_5 = -0.0030$ when the interaction between the pre-test and the group3 increases by one unit the Post-test average decreases by -0.0030 With other factors remaining constant.

The Hypotheses:

Interaction:

H_0 : There is no interaction effect between pre-test scores and group 2 membership on post-test scores.

H_1 : There is an interaction effect between pre-test scores and group 2 membership on post-test scores.

p\$-value for $\hat{\beta}_2 = 0.6623 > \alpha = 0.05$, we Do not reject H_0 at $\alpha = 0.05$, There is no interaction effect between pre-test scores and group 2 membership on post-test scores.

H_0 : There is no interaction effect between pre-test scores and group 3 membership on post-test scores.

H_1 : There is an interaction effect between pre-test scores and group 3 membership on post-test scores.

p\$-value for $\hat{\beta}_2 = 0.9784 > \alpha = 0.05$, we Do not reject H_0 at $\alpha = 0.05$, There is no interaction effect between pre-test scores and group 3 membership on post-test scores.

Compare models:

we use Anova to compare the two models

Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
41	6.940805	NA	NA	NA	NA
39	6.896055	2	0.0447504	0.1265408	0.8814985

$H_0: B_4=B_5=0.$

$H_a: B_4 \neq B_5 \neq 0$

$p - value = 0.8815 > \alpha = 0.05$, we do not reject H_0 at $\alpha = 0.05$, The models are equal or The intersection has no effect or importance.